

Probabilistic Electricity Price Forecasting under Regime Shift

Static and adaptive conformal calibration in a leakage-controlled simulated day-ahead benchmark

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Abstract

This project asks whether prediction intervals can respond when electricity-price behaviour changes. It uses a simulated hourly benchmark and keeps model fitting, calibration and evaluation in time order. A seasonal-naive forecast is the point baseline. Seven pooled gradient-boosting models, one for each quantile level and all conditioned on forecast horizon, estimate marginal uncertainty for 24 hourly targets. Raw intervals are compared with pooled static CQR, horizon-specific static CQR and adaptive CQR. In the final 83-day chronological evaluation block, raw 80% intervals covered 54.8% of observations. Adaptive CQR reached 82.6% coverage with a mean width of 22.50 EUR/MWh and the lowest observed interval score, 30.95. It still missed many extreme prices. These are point estimates from one fixed simulation seed, not employer or live-market results.

Data and contribution boundary All results come from the included simulator. The project uses established forecasting and conformal methods to build, test and question a reproducible prototype. It is not presented as a new algorithm or as evidence from a real electricity market.

1. Motivation and research question

My motivation came from work in the production-operations department of Tarim Oilfield's new-energy business. I compared photovoltaic forecast curves with actual output and saw that a gap could have several causes: weather, equipment availability, curtailment, maintenance or missing data. That experience made the problem concrete for me. It did not provide data for this portfolio. Here I ask whether an uncertainty interval can adapt after a change without becoming so wide that it stops being useful.

Choices behind the experiment

The rows are not randomly shuffled. A random split would let the model learn from the later renewable-shift period and then appear to predict an earlier version of the same period. That is not how a day-ahead system would be used.

The comparison starts with a seasonal-naive forecast and a tree-based quantile model. They are easier to inspect than a large neural network. This separates two questions: is the centre forecast better, and are the uncertainty intervals reliable? A more complex model can be tested later, after the evaluation pipeline is trusted.

2. Data and methods

2.1 Transparent simulated market

The simulator creates 10,080 hourly observations in three periods: stable operation, higher volatility and a renewable shift. Price depends on net load, scarcity, renewable surplus and changing noise. Load, renewable and temperature forecast proxies are created by perturbing simulated realised values with noise. This gives a controlled issue-time benchmark, but the proxies may be more informative than real operational forecasts. Short renewable outages and missing historical prices are also simulated. Missing prices are flagged and forward-filled; the pipeline stops rather than use future values if the required history starts with missing data. The true regime label is used only for evaluation.

Component	Implementation
Forecast setup	Seven pooled models, one per quantile, conditioned on horizon; 24 hourly targets
Past-only features	24h/168h price lags, trailing moments, missingness rate
Future-known proxies	Day-ahead load, renewable and temperature forecasts
Target	Hourly price in EUR/MWh

2.2 Models and calibration

Baseline: The seasonal-naive forecast uses the price at the same target hour one day earlier.

Quantile model: One pooled histogram gradient-boosting model is fitted at each of seven quantile levels from q05 to q95. Horizon is an input feature; the implementation does not fit 24 separate models. Estimated quantiles are sorted within each row if they cross.

Static CQR comparators: The pooled version uses one fixed correction for all 24 horizons. A second static version learns a separate fixed correction for each horizon. This separates horizon conditioning from online adaptation.

Adaptive CQR: Each forecast horizon uses its own recent error buffer and a changing local miscoverage rate. The interval is calculated before the relevant outcome is added, so an observation cannot change its own interval.

2.3 Validation and metrics

The final 83 issue days form a chronologically separated evaluation block and are not used for model fitting or conformal calibration. The model is also refitted at three earlier origins, with calibration based only on past data. Each origin evaluates raw, pooled static, horizon-specific static and adaptive intervals. MAE checks the median forecast; coverage, width, pinball loss, approximate CRPS and interval score provide complementary checks. The final block also supports a post-hoc gamma sensitivity diagnostic; it is therefore an evaluation period, not an untouched test set.

2.4 Relationship to existing work

Prior work already establishes CQR, adaptive conformal inference, day-ahead electricity-price applications and links to simulated battery trading (Romano et al., 2019; Gibbs and Candes, 2021; Zaffran et al., 2022; O'Connor et al., 2025). P1 therefore contributes a reproducible benchmark with static comparisons and failure analysis, not a first application or new algorithm.

Portfolio contribution The useful output is the full experiment: time-safe features, transparent baselines, pooled and horizon-specific calibration, stress tests, reproducible code and a record of where the methods fail. It is a research-training prototype, not a novel algorithm.

3. Results

Simulated-result notice Every number, table and figure in Sections 3-4 is generated from the simulated benchmark supplied with this repository; none is a Tarim Oilfield, employer or live-market result.

Interval	Coverage	Width	Interval score	Median MAE
Raw quantiles	0.548	11.76	34.35	6.30
Static CQR pooled	0.924	27.64	32.18	6.30
Static CQR by horizon	0.933	29.29	33.56	6.30
Adaptive CQR	0.826	22.50	30.95	6.30

The median model lowers MAE from 11.27 to 6.30 EUR/MWh. Raw intervals are too narrow after the shift. Pooled static CQR reaches 92.4% coverage with a width of 27.64. Horizon-specific static CQR reaches 93.3% coverage, with a width of 29.29 and an interval score of 33.56. Adaptive CQR reaches 82.6% coverage with a width of 22.50 and an interval score of 30.95. This is the lowest observed interval score among the four methods in this single simulation. It is not evidence that adaptation is generally better in real markets.

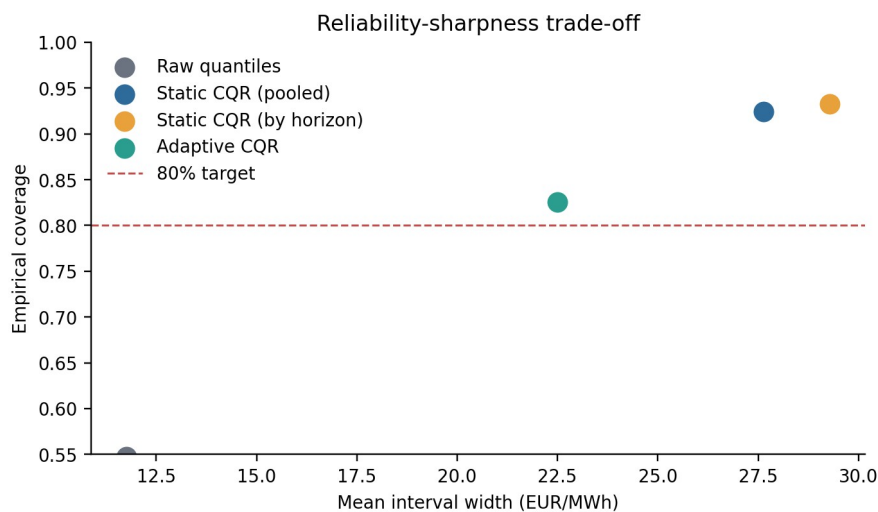


Figure 1. Reliability and sharpness in the chronological evaluation period for all four simulated interval methods; the dashed line is the 80% target.

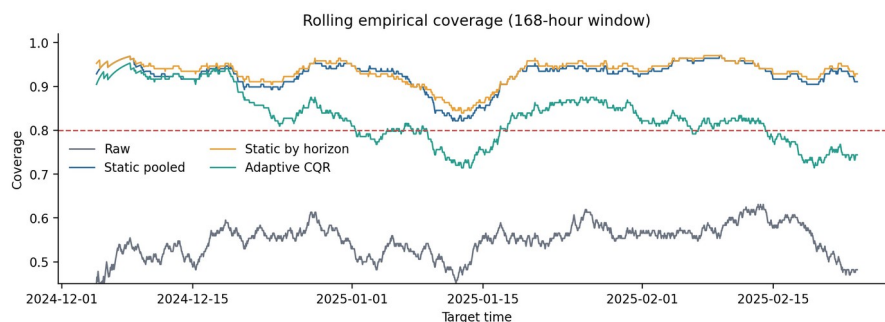


Figure 2. Rolling coverage of all four simulated interval methods after the renewable-shift boundary.

4. Stress tests, limitations and next steps

Adaptive-CQR coverage is 0.826 overall (n=1,992), 0.627 for price extremes (n=400) and 0.710 for the feature-distance OOD flag (n=131).

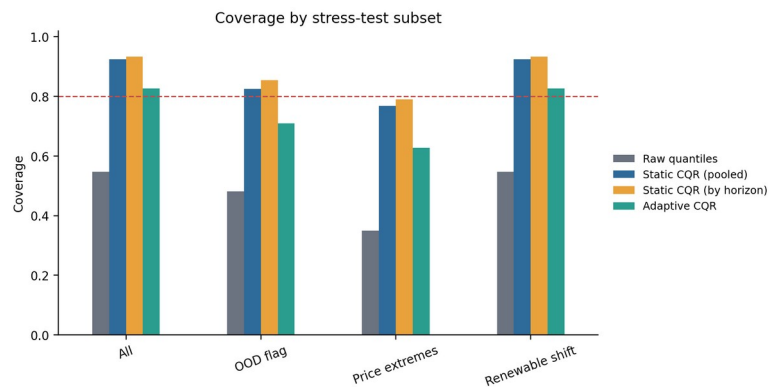


Figure 3. On the simulated benchmark, calibration helps overall reliability but price extremes remain a failure mode.

4.1 What failed

Adaptive CQR covers only 62.8% of extreme prices and 71.0% of feature-distance OOD observations. It controls a long-run error signal but cannot create information about rare spikes. In the three earlier rolling-origin folds, its coverage is 0.832, 0.747 and 0.839. The middle fold remains difficult. The saved rolling-origin table reports all four interval methods, so these adaptive results are not confused with static CQR.

4.2 Parameter sensitivity

A post-hoc sensitivity run increased the adaptive step size from 0.015 to 0.050. Coverage moved from 0.826 to 0.807, mean width fell from 22.50 to 21.30 EUR/MWh, and the interval score became slightly worse (30.95 to 31.07). Extreme-price coverage fell to 0.603. Both settings were assessed on the same evaluation block, so this diagnostic does not provide a clean basis for tuning gamma or selecting a winner. Gamma 0.015 is retained as the pre-specified reference configuration.

4.3 Main limitations in plain language

Results come from one fixed simulation seed, so there is no across-seed uncertainty estimate. Synthetic forecast proxies may be more informative than real forecasts. Pooled coverage may hide horizon-specific failures, and marginal hourly quantiles do not form reliable joint 24-hour paths. Adaptive calibration still misses spikes and OOD cases. Seven quantiles, one benchmark and a clipped online update cannot establish general superiority or exchangeable-data guarantees.

4.4 Reproducibility and next step

Isolated baseline run `python scripts/run_experiment.py --config configs/base.yaml --output-root evidence/runs/baseline`

The run manifest records seed 20260817, data fingerprint 6bacc0ff00f87704..., software versions and the full configuration. Exact versions are in requirements-lock.txt, and separate output roots prevent runs from overwriting one another. My next checks would repeat the experiment across several seeds, report uncertainty by day blocks, inspect coverage by horizon, and then move to publication-time-safe ENTSO-E, Elexon or OPSD data. A companion project examines how forecast distributions affect storage decisions.

References

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